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Section A: Definitions

- **Sleep Apnea:** A reduction in nasal airflow while sleeping for a duration more than the duration needed to complete 2 consecutive respiratory cycles/ rate according to age.
- **Obstructive sleep apnea :** a sleep-related breathing disorder/ apnea characterized by intermittent upper airway obstruction or collapse with presence of respiratory effort to breath, lasting more than 2 baseline respiratory cycles **with or without** an arousal or oxygen desaturation of at least 3% from baseline
- **Central apnea:** a sleep-related breathing disorder/ apnea characterized by the absence of chest and/or abdominal movement associated with a cessation of airflow lasting more than 2 baseline respiratory cycles and **must be** associated with an arousal or oxygen desaturation of at least 3% from baseline.
- **Nocturnal hypoventilation : evidence of carbon dioxide (CO₂) retention during sleeping which is defined as** pCO₂ level >50 mmHg for >25% of the total sleep time in the pediatric population (based on polysomnography test) or morning hypercapnia with pCO₂ in blood gas of > 50 mmHg.

Section B: Admission criteria for overnight pulse oximetry:

- **Purpose of admission:**
 1. Diagnosis by screening overnight pulse oximetry and morning blood gas
- **Eligible patients:**
 1. Patients with suspected sleep disordered breathing (SDB) who has Significant day and night Obstructive sleep apnea (OSA) symptoms , who are anticipated to have nocturnal hypercapnea and possible pulmonary hypertension as a complication of severe OSA (these patients need morning blood gas and ECHO
 2. Patients with suspected sleep disordered breathing (SDB) and one of the following:
 - Complex cardiac conditions
 - Genetic syndromes with dysmorphic features.
 - Chronic respiratory illnesses
 - Neuromuscular conditions with suspected nocturnal hypoventilation.
 - Infants who are less than 1 year of age

- 3- Any patient with suspected SDB based on symptoms in the absence of overnight pulse oximetry devices that are used for home studies where we will be obliged to admit them due to shortage of devices

- **Plan during admission:**

- **Admission place:**

1. High dependency (HD) admission:

- For patient who had previous history of significant and prolonged desaturation (persistent SpO₂ < 70 % lasting for more than 5 minutes) during sleep
- For patient with history of prolonged cyanotic spells during sleeping which indicate possible pulmonary hypertension crisis
- Patients with evidence of significant nocturnal hypoventilation on previous assessment (nocturnal pCO₂ >80 mmHg)

2. General wards (P1/P3/P4) :

- All patients who are not full filling the HD admission criteria

- **Admission notes and assessment:**

1. To be done by pulmonology team intern during working hours and by the intern on-call during on-call hours. (Note: Usually those patients come during working hours except for Saturday).
2. Patients should be seen by pulmonology team consultant/ registrar during working hours and by the registrar oncall after hours to make sure that the patient is clinically fit for test and that he does not need any investigations or frequent assessment.
3. The patient should be asymptomatic for any respiratory infection (upper or lower respiratory infections) to be fit for the test

- **Admission investigations :**

1. To request diagnostic pulse oximetry from procedure list and to call the on call respiratory therapist to reserve the machine and hook it to the patient at the nighttime
2. Overnight pulse oximetry should be done if possible, at room air, if any persistent desaturation noted because of obstruction while sleeping (below 90 %) continuous for more than 2 minutes, O₂ therapy can be given. Starting with 0.5 L of O₂/ minute and titrating up gradually and slowly if desaturation persists and consider looking for other causes of desaturation.
3. Routine blood investigation usually not needed on the day of admission

4. Intravenous access is not indicated because all of above-mentioned patient come from home for non-invasive test (overnight pulse oximetry) and their status usually reflect their baseline status at home
5. Morning blood gas to be done next morning before the patient wakes up or immediately (within 10 minutes) after waking up (could be capillary/ venous/ arterial blood gas)

Section C : Admission criteria for initiation of nocturnal non-invasive ventilation :

- **Purpose of admission:**
 - Initiation of nocturnal NIV (CPAP/BIPAP)
- **Eligible patients:**
 - Patients with confirmed sleep disordered breathing
- **Plan during admission:**
- **Admission place:**
 1. High dependency admission:
Patients with confirmed sleep disordered breathing plus one of the following criteria:
 - Severe OSA with Obstructive apnea hypopnea index (OAHI) > 20 events/hr (based on polysomnography report)
 - Severe OSA with McGill oximetry score equal > 3 (indicates severe OSA) based on overnight pulse oximetry.
 - Severe nocturnal hypoventilation with PCO₂ > 80 mmHg.
 - Complex cardiac conditions
 - Patients who failed nocturnal NIV training in general ward.
 1. General wards (P1/P3/P4) :
Patients with confirmed sleep disordered breathing and not full filling HD admission criteria
- **Admission note:**
 1. To be done by pulmonology team intern during working hours and by the intern on-call during on-call hours if general ward admission and by RMO on-call if HD admission. (Note: Usually those patients come during working hours except for Saturday).
 2. Patients should be seen by pulmonology team consultant/ registrar during working hours and by the registrar on call after hours to make sure that
 - The patient is clinically fit to initiate nocturnal NIV (not acutely ill)
 - He is allocated in appropriate bed (general bed vs HD bed)
 - He does not need any investigations or frequent assessment if he is in general ward.

- **Admission plan:**
 1. To contact the pulmonologist in service/ on-call if the plan of the admission (starting settings of NIV, appropriate interface/mask and the need for any investigations) is not clear or not documented in the admission order.
 2. To contact the respiratory therapist on call for family orientation and education about the device, to check interface fitness and setting up the devices.
 3. To update the Pulmonologist in service about the patient progress on NIV, result of requested investigation (pulse oximetry report and morning blood gas) to decide if any NIV settings adjustments are required.

- **Admission investigations:**
 1. If the patient tolerates NIV for more than 4 hours continuously), a request for diagnostic pulse oximetry should be done the night after.
 2. Overnight pulse oximetry can be requested from procedure list under respiratory procedure section
 3. To call the on call respiratory therapist to reserve the machine and hook it to the patient at the nighttime when he is using his NIV devices.
 4. Routine blood investigation usually not needed on the day of admission
 5. Intravenous access is not indicated (except for HD patients): Because all of above mentioned patient come from home for initiation of noninvasive devices and their status usually reflect their baseline status at home
 6. Morning blood gas to be done next morning before the patient wakes up or immediately after waking up (could be capillary/ venous/ arterial blood gas)
 7. Morning blood gas should be repeated whenever the settings of NIV are increased unless the patient has normal pCO₂ (< 50 mmHg)
 8. If the patient NIV device has feature of storing the data in the machine, an attempt to look/ download this device should be done to help in deciding the need of adjusting the setting.

Section D: Admission criteria for titration of nocturnal non-invasive ventilation :

- **Purpose of admission:**
 - Titration (up or down) of nocturnal NIV (CPAP/BIPAP)

- **Eligible patients:**
 - Patients with confirmed sleep disordered breathing who are on nocturnal NIV

- **Plan during admission:**

- **Admission place:**
 1. General wards (P1/P3/P4) :
 - Patients with confirmed sleep disordered breathing who are already using nocturnal NIV at home
 - Patient should bring his own NIV device and mask on day of admission

- **Admission note:**

1. To be done by pulmonology team intern during working hours and by the intern on-call during on-call hours (Note: Usually those patients come during working hours except for Saturday).
2. Patients should be seen by pulmonology team consultant/ registrar during working hours and by the registrar on call after hours to make sure that
 - The patient is clinically fit for titration study (not acutely ill)
 - He is allocated in appropriate bed
 - He does not need any investigations or frequent assessment if he is in general ward.
 - To make sure that the patient has brought his own NIV device and mask he is using at home.

- **Admission plan:**

1. To contact the pulmonologist in service/ on-call if the plan of the admission (overnight pulse oximetry on patient current NIV setting and the need for any investigations) is not clear or not documented in the admission order.
2. To contact the respiratory therapist on call, to check the settings of the device for documentation purposes, to check interface fitness and for hooking overnight pulse oximetry while the patient on NIV during sleeping.
3. To update the Pulmonologist in service about the patient progress on NIV, result of requested investigation (pulse oximetry report and morning blood gas) to decide if any NIV settings adjustments are required.

- **Admission investigations:**

9. Overnight pulse oximetry can be requested from procedure list under respiratory procedure section
10. To call the on call respiratory therapist to reserve the machine and hook it to the patient at the nighttime when he is using his NIV devices.
11. Routine blood investigation usually not needed on the day of admission
12. Intravenous access is not indicated Because all of above mentioned patient come from home for titration study of NIV devices and their status usually reflect their baseline status at home
13. Morning blood gas to be done next morning before the patient wakes up or immediately after waking up (could be capillary/ venous/ arterial blood gas)
14. If the patient NIV device has feature of storing the data in the machine, an attempt to look/ download this device should be done to help in deciding the need of adjusting the setting.
15. Morning blood gas should be repeated whenever the settings of NIV are increased unless the patient has normal pCO₂ (< 50 mmHg) before.